MILITARY SPECIFICATION

GASKETS, METALLIC-ASBESTOS, SPIRAL WOUND

This amendment forms a part of Military Specification MIL-G-0021032E(SHIPS), dated 6 September 1974.

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* 2.1, under "SPECIFICATIONS, FEDERAL" add the following:

"QQ-C-320 - Chromium Plating (Electrodeposited)."

* 2.1, under "SPECIFICATIONS, MILITARY" add the following:

"MIL-P-17667 - Paper Wrapping, Chemically Neutral (Non-Corrosive).
MIL-B-22191 - Barrier Materials, Transparent, Flexible, Heat Sealable."

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* 3.1.1.1, line 9: Delete "except that y be modified to read minimum 6000 psi."

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* 3.1.1.2, line 8: Delete "and as modified in 3.1.1.1.".

* 3.1.2, line 3: Delete "0.125 $\pm$ 0.006" and substitute "0.125 $^{+0.006}_{-0.008}$".

* 3.1.2, line 6: Delete "0.119" and substitute "0.117".

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* Table III, under columns "Metal outer ring o.d. inch" and "Test load pounds" delete "Series 900".

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* 3.1.2.2, line 2: Add the following: "Initial gasket thickness limits apply to the measurement of the metal portion of the gasket and do not include asbestos filler material which may protrude slightly when not compressed."

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* 3.2.2, line 2: Delete "of soft carbon" and substitute "of carbon".

* 3.2.2, line 4: Add the following: "When specified (see 6.1), the carbon steel outer ring shall be chromium plated in accordance with QQ-C-320. The use of coatings composed of metallic materials having melting points lower than the maximum allowable gasket service temperature (see 1.2.1) is prohibited. The use of cadmium plating is prohibited."

* 3.3.5, add the following new paragraph:

"3.3.5 Surfaces and edges of gaskets and metal outer rings shall be free of burrs and loosely adhering metal."

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* 3.4.3, line 5: Delete "stapled" and substitute "sealed".

* 3.5 and 3.6, add the following new paragraphs:

"3.5 Cleanliness requirements. Surface of gaskets and related parts shall be commercially clean, i.e., free of dirt, oils, grease, chips, and any foreign matter.

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"3.6 Production equipment inspection. An inspection shall be performed prior to starting a production run of MIL-type gaskets to assure that equipment and process lines have been purged of material from the previous production run. The inspection shall be performed on all equipment used in manufacturing operations where the material is not segregated and positively identified."

* Table VI, delete and substitute:

"Table VI - Sampling for visual, dimensional, and alloy examination.

Lot size (number of gaskets)	Sample size (number to be examined)	Acceptance number 1/ (defective)	Rejection number 1/ (defective)
2 to 10	All	0	1
11 to 25	10	0	1
26 to 40	15	0	1
41 to 110	25	1	2
111 to 300	35	1	2
301 to 500	50	2	3
501 to 800	75	3	4
801 to 1300	110	4	5
1301 and over	150	5	6

1/ For visual and dimensional examination only."

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* Table VIII, add the following:

"109	Surfaces or edges of gaskets or metal rings having burrs or loosely adhering metal.
110	Surfaces of gaskets or related parts exhibiting dirt, oils, grease, chips or any foreign matter."

* 4.3.1, lines 3 and 4: Delete "20 percent of the width of the metal strip" and substitute "the following dimensions:

Gasket thickness	Axial direction (see figure 1A)	Circumferential direction (see figure 1A)
0.125 inch	0.034 inch	0.102 inch
0.175 inch	0.045 inch	0.134 inch

Acceptance criteria for defective spot welds shall be as specified in table IX."

* After paragraph 4.3.1, insert the following as table IX:

"Table IX - Examination of spot welds. 1/

Number of spot welds in gasket I.D.	Acceptance number (defective)	Rejection number (defective)
3 to 4	0	1
5 to 7	1	2
8 to 11	2	3
12 to 15	3	4
16 and over	4	5

1/ Any gasket, regardless of number of spot welds, containing 2 defective welds adjacent to each other shall be considered a reject."

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* 4.4.1.2, delete second sentence and substitute: "Where a finer finish is required and so specified (see 6.1), plates with a roughness range of 63 to 125 RHR shall be used."

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• 4.4.2, delete and substitute:

"4.4.2 Alloy identity. Prior to packaging, each of the sample gaskets selected in accordance with 4.2.2 or, when specified (see 6.1), each of the gaskets in the lot, shall be examined for alloy identity by a spot tester manufactured by Systems Scientific Laboratory, Ridgewood, New Jersey 07451, or Spot Testers Inc., Springfield, New Jersey 07081, or other equal tester, which has been demonstrated to the satisfaction of the Government inspection personnel. Test plates (for calibration purposes) of the alloys to be tested shall be provided for each tester. If any sample fails the alloy identity examination, the entire lot shall be rejected."

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• 4.6, line 1, at beginning of first sentence, add: "Unless otherwise specified (see 6.1),".

• 5.1.1.1, delete first sentence and substitute: "When specified (see 6.1), gaskets shall be bundled, tied, and wrapped in transparent barrier material conforming to MIL-B-22191 or chemically neutral wrapping paper conforming to MIL-P-17667."

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• 5.2.4, delete second and third sentence and substitute: "Loose fill materials shall be non-corrosive to type 347 or 309S corrosion-resisting steel, nickel-chromium-iron alloy, nickel-copper alloy, or carbon steel. Loose excelsior, shredded newspapers, and similar materials that have not been refined for noncorrosiveness to these alloys shall not be used. Use of polystyrene "loose-fill" material is prohibited."

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• 6.1, add the following:

"(l) When chromium plating of metal outer ring is required (see 3.2.2).
(m) When finer surface finish is required for steel test plates (see 4.4.1.2).
(n) When each of the gaskets in a lot shall be examined for alloy identity (see 4.4.2).
(o) When certification of quality conformance is not required (see 4.6).
(p) When MIL-B-22191 or MIL-P-17667 packaging material is required (see 5.1.1.1)."

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Figure 1: Delete "1/8" ± .006" and substitute "0.125 $^{+.006}_{-.008}$ inch".

• After page 14, add the following as figure 1A.

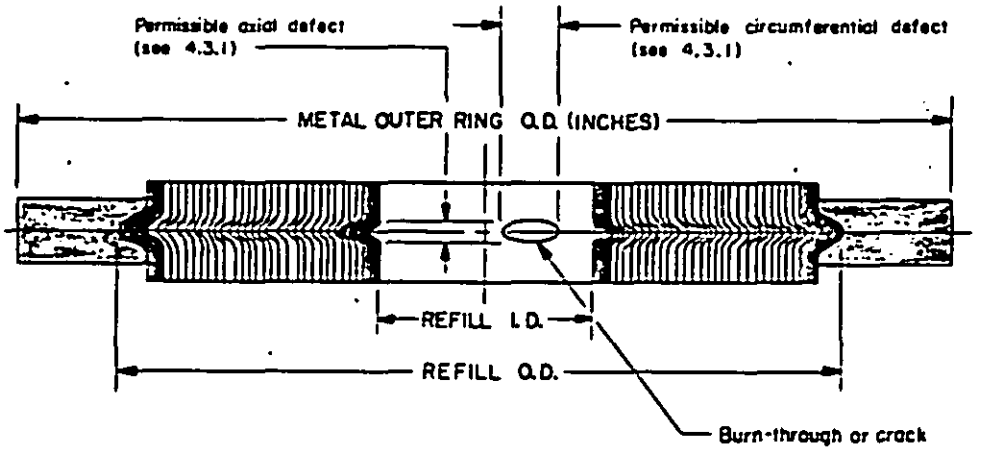


Figure 1A - Gasket, spot weld examination.

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* Figure 2, at bottom of page, delete certification statement and substitute:

"We hereby certify that the metal strip used in the spiral wound metallic-asbestos gaskets is in accordance with the specification listed above. (Submittal of actual test data for the metal strip is not required.) We further certify that the gaskets have been tested in accordance with MIL-G-21032 as modified by the applicable amendments. We certify, also, that the gaskets have not come in contact with mercury or any of its compounds."

NOTE: The margins of this amendment are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

Preparing activity:
Navy - SH
(Project 5330-N060)

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